

TURNKEY LINUX ACTIVE DIRECTORY 18.1 DOMAIN CONTROLLER & DEBIAN CLIENT CONFIGURATION GUIDE

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LINUX CLIENT CONFIGURATION

Install Required Packages

```
# sudo apt install -y realmd sssd sssd-tools libnss-sss libpam-sss krb5-user adcli samba-common-bin oddjob oddjob-mkhomedir
```

Enable Make Home Directory For AD User On First Login

```
# sudo pam-auth-update --enable mkhomedir
```

Download A Copy Of The OpenRSAT Application

OpenRSAT is a very good AD management tool. Because it is GTK2 it can have some weird rendering characteristics. Often position your cursor on the window title bar and F5 to reload.

<https://github.com/tranquilit/OpenRSAT>

Add / Edit /etc/krb5.conf On The Linux Client & chmod **644**

Set DOMAIN.LAN to match your domain name:

```
[libdefaults]
```

```
default_realm = DOMAIN.LAN
```

```
dns_lookup_realm = false
```

```
dns_lookup_kdc = false
```

```
[realms]
```

```
DOMAIN.LAN = {
```

```
    kdc = dc1.domain.lan
```

```
    admin_server = dc1.domain.lan
```

```
}
```

```
[domain_realm]
```

```
.domain.lan = DOMAIN.LAN
```

```
domain.lan = DOMAIN.LAN
```

Add AD Server IP To /etc/hosts - DNS Is NOT Actually Required!

Add a line to resolve your server name to /etc/hosts:

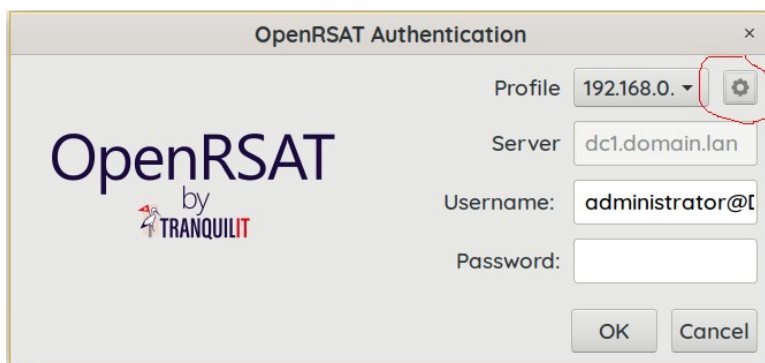
```
192.168.0.200 dc1 dc1.domain.lan domain.lan
```

Join The Domain

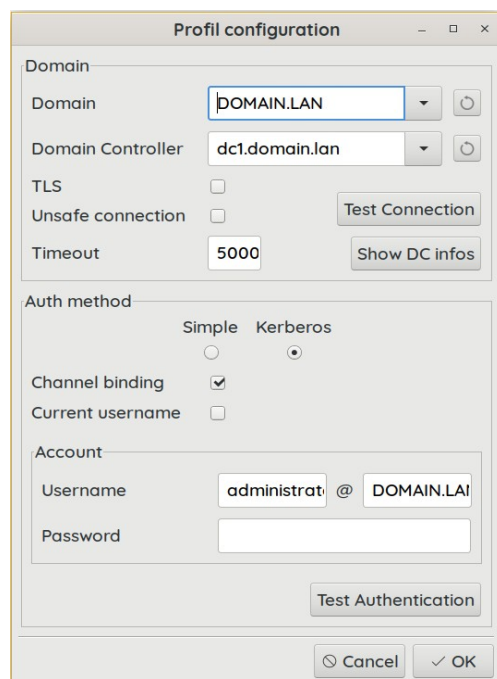
```
# sudo realm join -U administrator domain.lan
```

Configure OpenRSAT To Connect To The AD Controller

Click the gear icon to add a new profile:



Add the connection parameters:



Create An AD User & Group

Use OpenRSAT, webmin: <https://xxx.xxx.xxx.xxx:12321> samba module, or samba-tool in the ssh/terminal.

- On the AD server create a new group "[linuxlogins@domain.lan](#)"
- Create domain login users and add them to the group.

Give Permission To The AD Group To Login To The Linux Client

```
# sudo realm permit -g "linuxlogins@domain.lan"
```

How To Leave The Domain

```
# sudo realm leave --remove -U administrator domain.lan
```

There could be times you want to flush SSSD cached credentials to be sure - using these commands:

```
# sudo rm -rf /var/lib/sss/db/*  
# sudo systemctl restart sssd  
# sudo rm -f /etc/sss/sss.conf  
# sudo systemctl restart realmd
```

SERVER CONFIGURATION

Configure Winbind So AD Can Be Referenced By chown:chrgp

Edit file /etc/nsswitch.conf and add winbind:

```
passwd:    files winbind
group:     files winbind
```

Install the libnss-libwind package:

```
# sudo apt-get install libnss-winbind
# ln -sf /usr/lib/x86_64-linux-gnu/libnss_winbind.so.2 /lib/x86_64-linux-gnu/libnss_winbind.so
# ldconfig
# systemctl restart winbind
# systemctl restart samba-ad-dc
```

Turnkey Default Shipped Samba Share Configuration (smb.conf)

This is the out-of-the-box Turnkey configuration (if you ever want to revert & start over):

BTW ... this uses NTLM weak security, and does not enforce encryption.

Any client at all on the network can authenticate using Network Neighbourhood to locate the shares, simply to be prompted for WORKGROUP credentials.

Domain membership is not enforced in this configuration.

[global]

```
dns forwarder = 8.8.8.8
interfaces = 127.0.0.1 192.168.0.200 2001:8003:2762:ee00:2ed0:5aff:fe17:3365
netbios name = DC1
realm = DOMAIN.LAN
server role = active directory domain controller
workgroup = DOMAIN
idmap_ldb:use rfc2307 = yes
```

Same with smb3 mininum protocol required and force encyption added:

[global]

```
dns forwarder = 8.8.8.8
interfaces = 127.0.0.1 192.168.0.200 2001:8003:2762:ee00:2ed0:5aff:fe17:3365
netbios name = DC1
realm = DOMAIN.LAN
server role = active directory domain controller
workgroup = DOMAIN
idmap_ldb:use rfc2307 = yes
```

```
server smb encrypt = yes  
server min protocol = SMB3
```

AD Secure Kerberos Authenticated Samba With SMB3 & Encryption

Domain Membership & AD Login Is **Enforced** In This Configuration.
This is probably what you want in an enterprise scenario:

[global]

```
dns forwarder = 8.8.8.8  
interfaces = 127.0.0.1 192.168.0.200  
netbios name = DC1  
realm = DOMAIN.LAN  
server role = active directory domain controller  
workgroup = DOMAIN  
idmap_ldb:use rfc2307 = yes  
map to guest = Never  
restrict anonymous = 2  
server smb encrypt = yes  
server min protocol = SMB3  
ntlm auth = disabled  
client use kerberos = required
```

CONFIGURING A HOME SHARE & PERMISSIONS

Create the home share folder on the linux AD server:

```
# mkdir -p /srv/share/home  
# chown nobody:'DOMAIN\domain users' /srv/share/home  
# chmod 050 /srv/share/home
```

Add this share definition to /etc/smb/conf:

```
[home]  
comment = Active Directory User Home Folders  
path = /srv/share/home/  
writeable = yes  
valid users = @"domain users"  
map archive = no  
map system = no  
map hidden = no  
inherit permissions = yes  
create mask = 0660  
directory mask = 0770  
force create mode = 0660  
force directory mode = 0770  
inherit owner = yes  
vfs objects = acl_xattr  
map acl inherit = yes  
store dos attributes = yes  
guest ok = no
```

Create a home sub-folder per user and restrict access to the user only:

```
# mkdir /srv/share/home/username  
# chown 'DOMAIN\<AD username>':'DOMAIN\domain users' /srv/share/home/username  
# chmod 700 /srv/share/home/<username>
```

**** You could change the group to be a backup operators group later and set chmod 750.

CONFIGURING A GROUP SHARE WITH SUB-FOLDERS

Create the group share folder on the linux AD server:

```
# mkdir -p /srv/share/groups  
# chown nobody:'DOMAIN\domain users' /srv/share/groups  
# chmod 550 /srv/share/groups
```

Add this share definition to /etc/smb/conf:

```
[groups]  
comment = Active Directory Group Folders  
path = /srv/share/groups/  
writeable = yes  
valid users = @"domain users"  
map archive = no  
map system = no  
map hidden = no  
inherit permissions = yes  
create mask = 0660  
directory mask = 0770  
force create mode = 0660  
force directory mode = 0770  
inherit owner = yes  
vfs objects = acl_xattr  
map acl inherit = yes  
store dos attributes = yes  
guest ok = no
```

Use OpenRSAT, Webmin or samba-tool to create an AD group for example “MARKETING”.

To create a sub-folder per group and restrict access to that group using chown/chmod:

```
# mkdir /srv/share/groups/marketing  
# chown nobody:'DOMAIN\MANAGEMENT' /srv/share/groups/marketing  
# chmod 2770 /srv/share/group/marketing
```


Create a sub-folder for multiple groups using combination chown/chmod & Posix ACL's

```
# mkdir /srv/share/groups/transport
```

```
# chown nobody:"DOMAIN\domain users" /srv/share/groups/transport
```

```
# chmod 2700 /srv/share/groups/transport
```

```
# setfacl -m g:"DOMAIN\transport":rw /srv/share/groups/transport
```

```
# setfacl -m g:"DOMAIN\marketing":rw /srv/share/groups/transport
```

**** *You can also use ACL's to give backup operators and groups access*

Listing POSIX ACL'S

```
getfacl -d file/foldername
```

Remove an individual ACL

```
setfacl -x g:groupname file/folderpath
```

Remove all ACL'S

```
setfacl -b file/folderpath
```

To prevent users from deleting a home or group sub-folder apply an immutable hidden file:

```
# > /srv/share/home/<AD username>/.deletion_prevented
```

```
# > chattr +i /srv/share/home/<AD username>/.deletion_prevented
```

```
# > /srv/share/groups/<AD groupname>/.deletion_prevented
```

```
# > chattr +i /srv/share/groups/<AD groupname>/.deletion_prevented
```